

DATA REPRESENTATION PRACTICE SHEET

NIMCET TARGET PAPER • INSANE DIFFICULTY LEVEL

ResourceOS

1. In an 8-bit floating-point system with 1 sign bit, 4 exponent bits (bias = 7), and 3 fraction bits, what exact decimal value is represented by the bit pattern 0 0000 011 (a subnormal number)?

- (A) $3 / 256$ (B) $3 / 512$
(C) $3 / 1024$ (D) $3 / 128$

2. Consider a base- r integer system. If the $(r - 1)$'s complement of an n -digit integer N with fractional part of m digits is defined precisely, what is the exact formula for the r 's complement of N ?

- (A) $r^n - N$ (B) $r^n - r^{-m} - N$
(C) $r^n - N + r^{-m}$ (D) $(r^n - r^{-m}) - N + 1$

3. Convert decimal -23 into its equivalent 8-bit Negabinary (Base -2) representation.

- (A) 11010001 (B) 11101001
(C) 11000101 (D) 11111001

4. In IEEE 754 single-precision addition using Guard (G), Round (R), and Sticky (S) bits, what are the values of G, R, S before rounding when adding $1.000000000000000000000000 \times 2^0$ and $1.101100000000000000000000 \times 2^{-25}$?

- (A) G=0, R=0, S=1 (B) G=1, R=1, S=0
(C) G=0, R=1, S=1 (D) G=0, R=0, S=0

5. In an Extended Hamming (8,4) SEC-DED code, parity bit P_8 covers all bits 1 through 8. If the received word has syndrome bits $S_4S_2S_1 = 101$ and overall parity P_{overall} indicates an error, what type of error occurred?

- (A) Single-bit error at position 5 (B) Double-bit error detected
(C) Single-bit error at position 3 (D) Parity bit P_8 itself is corrupted

6. If $(123)_b + (234)_b = (402)_b$ in a positional number system with base b , what is the value of b ?

- (A) 5 (B) 6
(C) 7 (D) 8

7. What is the smallest positive normalized number in IEEE 754 single-precision format expressed in exact mathematical form?

- (A) 2^{-126} (B) 2^{-127}
(C) $(1 - 2^{-23}) \times 2^{-126}$ (D) 2^{-149}

8. In a Q4.4 signed fixed-point format (1 sign bit, 3 integer bits, 4 fractional bits using 2's complement), what binary pattern represents decimal -3.625?

- (A) 11000110 (B) 11000111
(C) 10110110 (D) 11000101

9. Which boolean logic equation correctly detects overflow (V) in n -bit 2's complement addition $A + B = S$, where A_{n-1} , B_{n-1} , and S_{n-1} are the sign bits?

- (A) $V = (A_{n-1} \text{ AND } B_{n-1} \text{ AND NOT } S_{n-1}) \text{ OR } (\text{NOT } A_{n-1} \text{ AND NOT } B_{n-1} \text{ AND } S_{n-1})$ (B) $V = (A_{n-1} \text{ XOR } B_{n-1}) \text{ AND } S_{n-1}$
(C) $V = A_{n-1} \text{ XNOR } B_{n-1} \text{ XNOR } S_{n-1}$ (D) $V = C_{in} \text{ AND } C_{out}$

10. In a 12-bit Excess-2048 code system for representing signed integers, what decimal value does the binary pattern 011111111111 represent?

- (A) -1
- (B) 0
- (C) +2047
- (D) -2048

11. Calculate the smallest positive subnormal number in IEEE 754 single-precision floating-point format.

- (A) 2^{-149} (approx. 1.4×10^{-45})
- (B) 2^{-126}
- (C) 2^{-127}
- (D) 2^{-150}

12. What phenomenon occurs when subtracting two nearly equal floating-point numbers in single precision, leading to severe loss of significant digits?

- (A) Catastrophic Cancellation
- (B) Subnormal Underflow
- (C) Mantissa Overflow
- (D) Exponent Drift

13. If an n -bit unsigned integer is multiplied by $2^k - 1$ (where $k < n$), how can this operation be expressed using minimal logic shifts and subtractions?

- (A) $(X \ll k) - X$
- (B) $(X \ll (k - 1)) + X$
- (C) $(X \ll k) + X$
- (D) $(X \gg k) - X$

14. Convert decimal fraction 0.1_{10} to base-16 (hexadecimal) floating representation up to 4 hexadecimal places.

- (A) $0.1999..._{16}$
- (B) $0.1A3F..._{16}$
- (C) $0.186A..._{16}$
- (D) $0.2333..._{16}$

15. How many distinct pairs of numbers (A, B) in 4-bit 2's complement arithmetic cause an overflow condition upon addition $A + B$?

- (A) 56
- (B) 64
- (C) 42
- (D) 28

16. What is the hexadecimal bit pattern for Quiet NaN (qNaN) with a positive sign bit in IEEE 754 single precision, adhering to the standard default payload?

- (A) 0x7FC00000
- (B) 0x7F800001
- (C) 0xFFC00000
- (D) 0x7F800000

17. In a base-3 (ternary) system, what is the 3's complement of the integer $(2102)_3$?

- (A) $(0121)_3$
- (B) $(0120)_3$
- (C) $(0122)_3$
- (D) $(1011)_3$

18. What is the dynamic range (ratio of maximum representable magnitude to minimum non-zero magnitude) of an 8-bit unsigned integer system compared to an 8-bit sign-magnitude system?

- (A) 255 : 127
- (B) 256 : 128
- (C) 255 : 255
- (D) 512 : 256

19. Which polynomial generator $G(x)$ is standard for CRC-8 error detection algorithm?

- (A) $x^8 + x^2 + x + 1$ (B) $x^8 + x^4 + x^3 + 1$
(C) $x^8 + x^7 + x^3 + x + 1$ (D) $x^8 + 1$

20. Compute the exact difference between IEEE 754 single-precision float representations of 2.0 and the next larger representable float.

- (A) 2^{-22} (B) 2^{-23}
(C) 2^{-24} (D) 2^{-21}

21. If an n -bit 2's complement number contains k leading 1s, right shifting it arithmetic-wise by m positions (where $m < n$) guarantees:

- (A) The sign bit remains 1 and m 1s are prepended at MSB. (B) m 0s are prepended at MSB.
(C) Value is multiplied by 2^m . (D) Underflow error flag is generated.

22. In a custom floating-point format with 1 sign bit, 5 exponent bits (bias = 15), and 4 mantissa bits, what decimal value does 1 10010 1100 represent?

- (A) -14.0 (B) -28.0
(C) -7.0 (D) -56.0

23. What is the Hamming distance between the ASCII characters 'A' (01000001) and 'a' (01100001)?

- (A) 1 (B) 2
(C) 3 (D) 4

24. Adding two Excess-3 coded numbers requires a correction step. If the 4-bit addition of two digits does NOT produce an unadjusted binary carry out, what correction must be applied to that nibble?

- (A) Subtract 0011 (3) (B) Add 0011 (3)
(C) Add 0110 (6) (D) Subtract 0110 (6)

25. In 1's complement arithmetic, adding 11111110 (-1) to 11111101 (-2) in an 8-bit representation yields which uncorrected sum and final carry bit?

- (A) Uncorrected: 11111011, Carry: 1 (Final sum: 11111100 = -3) (B) Uncorrected: 11111011, Carry: 0 (Final sum: 11111011 = -4)
(C) Uncorrected: 11111100, Carry: 1 (Final sum: 11111101 = -2) (D) Overflow flag set, result invalid

ANSWER KEY

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	(B)	6	(A)	11	(A)	16	(A)	21	(A)
2	(B)	7	(A)	12	(A)	17	(A)	22	(A)
3	(A)	8	(A)	13	(A)	18	(A)	23	(A)
4	(A)	9	(A)	14	(A)	19	(A)	24	(A)
5	(A)	10	(A)	15	(A)	20	(A)	25	(A)